Collaborative Attention-Based Heterogeneous Gated Fusion Network for Land Cover Classification

Xiao Li, Lin Lei, Yuli Sun, Ming Li, and Guangyao Kuang, *Senior Member, IEEE*

***Abstract*— Existing land cover classification methods mostly rely on either the optical or synthetic aperture radar (SAR) features alone, which ignore the mutual complementary effects between optical and SAR sources. In this article, we compare the distribution histograms of deep semantic features extracted from optical and SAR modalities within land cover categories, which intuitively demonstrates that there are the large complementary potentials between the optical and SAR features. Therefore, we propose a novel collaborative attention-based heterogeneous gated fusion network (CHGFNet), which hierarchically fuses both optical and SAR features for land cover classification. More specifically, the CHGFNet consists of three main components: two-stream feature extractor, multimodal collaborative attention module (MCAM), and the gated heterogeneous fusion module (GHFM). Given optical and SAR patch pairs, two-stream feature extractor introduces multistage feature learning methodology to acquire discriminative optical and SAR features. Then, to explore the inherent complementarity between optical and SAR features, MCAM is embedded into CHGFNet, which provides an efficient stage to capture the correlation between optical and SAR features by jointly calculating the collaborative attention in joint feature space. Finally, to automatically learn the varying contributions of both optical and SAR features for classifying different land categories, GHFM is used to fuse both optical and SAR features. Extensive comparative evaluations demonstrate the advantages of CHGFNet within land cover classification over the state-of-the-art methods on three co-registered optical and SAR data sets.**

***Index Terms*— Collaborative attention model, complementary information, convolutional neural networks (CNNs), gated fusion, land cover classification.**

I. Introduction

With rapid development of earth observation technologies, various remote sensing (RS) resources have been widely considered in RS information processing [1]–[5]. Multimodal RS data from the same region have become more accessible, which can provide a variety of complementary information about land properties on land cover classification, such as the spectral information contained in optical images and the scattering information [6], [7] provided by synthetic aperture radar (SAR) images. It shows a promising opportunity to improve the land cover classification performance by taking full advantage of the complementary between optical and SAR modalities [8], [9].

The major challenges in the fusion of both optical and SAR modalities can be contributed to two aspects [10]–[13]. First, the semantic gap between the raw images and class labels is hard to span [10], [11]. Traditional hand-crafted feature extraction mechanism inevitably makes them unable to fully reflect the land cover information of complex scenes. Recently, deep convolutional neural networks (DCNNs) as the powerful feature extractors have achieved superior performance in the computer vision tasks [14], [15], which use the layer-by-layer nonlinear transformation to intelligently extract deep semantic features from raw images, and then relieve the semantic gap between the raw images and class labels. Then, due to imaging mechanisms between the optical and SAR images are radically different, there is a semantic gap between optical and SAR features, therefore, how to design the reasonable hetero-modal fusion strategy that establishes the sufficient interactions between optical and SAR features and unify the optical and SAR features, which becomes the key issues to be addressed.

Recently, collaborative attention embedded into the deep neural networks [16], [17] has been studied, which is attempted to apply in a range of computer vision tasks, such as person reidentification [18], [19], visual question answering [16], [20], [21] and hashtag recommendation [17], [22], [23]. In the person reidentification, collaborative attention mechanism has been used to correlate the feature maps corresponding to paired images and render respective features contextual aware of each other [18]. In the visual question answering field, collaborative attention mechanism has been used to jointly reason about the image and question attentions [20]. Given the representations of an image and a question, it generated an attention map of image regions for each question word, while it also generated an attention map of question words for each image region [21]. In the hashtag recommendation, the texts and the corresponding images of tweets provided the complementary information of each other [23], and the collaborative attention mechanism could incorporate the textual and visual information to recommend hashtags for multimodal tweets, which was helpful to improve the accuracy of recommendation. In a word, the collaborative attention mechanism can capture the semantic interaction between different modalities by learning the pairwise attention between two modalities. In this article, to explore the inherent complementarity among optical and SAR features, we embed multimodal collaborative attention module (MCAM) into the

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The authors are with the State Key Laboratory of Complex Electromagnetic Environment Effects on Electronics and Information System, National University of Defense Technology, Changsha 410073, China (e-mail: lxcherishm@126.com; alaleilin@163.com; sunyuli@mail.ustc.edu.cn; liming17@nudt.com; kuangyeats@hotmail.com).

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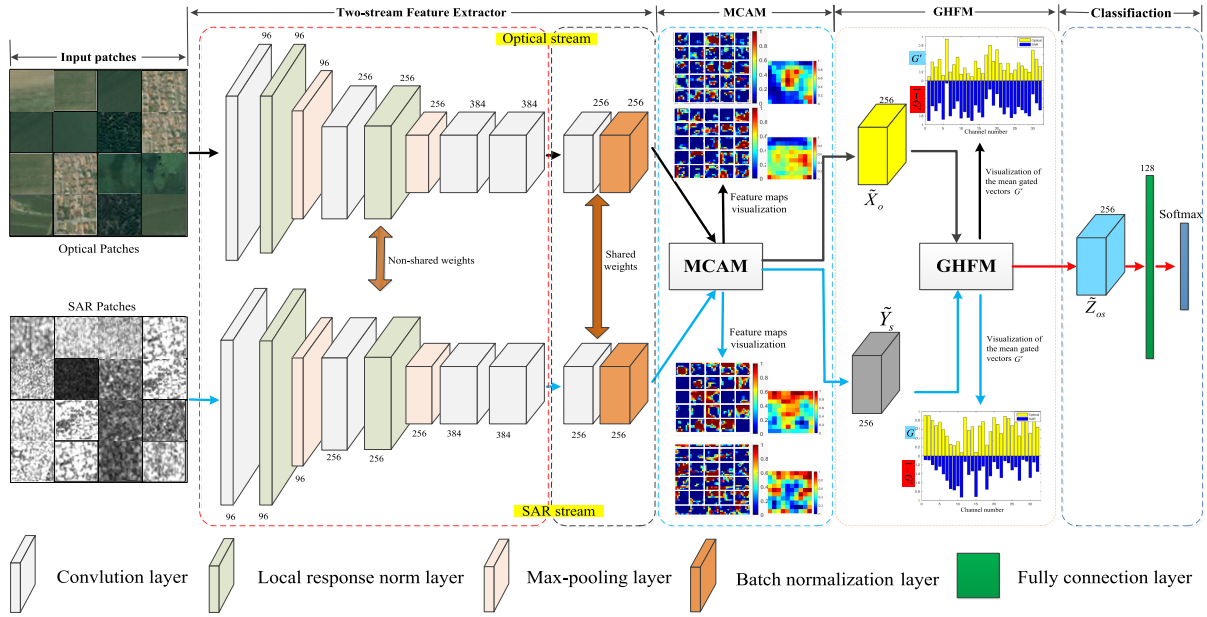


Fig. 1. Overall architecture of our CHGFNet. CHGFNet consists of four parts: 1) the front-end two-stream network as feature extractor; 2) the proposed MCAM. It can capture the semantic interaction between different modalities; 3) the GHFM, which learns the gate vectors to weigh the contribution of optical and SAR modalities for obtaining the fusion features; and 4) the classification. The cross-entropy is used as loss function.

proposed land cover classification framework, which provides an efficient stage to capture the correlation between optical and SAR features by jointly computing collaborative attention response into a joint feature space, and precisely focus on the important areas of feature maps.

Moreover, in most of the state-of-the-art deep fusion frameworks, linear fusion models (e.g., feature concatenation, element-wise production, and element-wise sum) are frequently utilized schemes for multimodal feature fusion, and they obtained good experimental results [24]–[26]. However, these basic fusion strategies are simple score fusion with equal weights for multiple modalities, which ignore the varying contributions of the modalities for classifying different land categories [25], [26]. It is necessary to propose reasonable fusion strategy for automatically learning different contributions of the optical and SAR features for improving land cover classification. Recently, the gated mechanism [27]–[32] has been received increasing attention in DCNN, which is used to control the propagation of useful information from different modalities. Gated mechanism provided an elegant fusion strategy to selectively fuse features from multiple levels for semantic segmentation in the gated fully fusion network (GFFNet) [29]. The gated bi-directional network (GBD-Net) [30] effectively integrated the local and contextual visual cues of image regions by gated mechanism for improving object detection. Highway network [32] used the gated mechanism to decrease the flow path of information through a network, which made it possible to train very deep networks. Inspired by the above works, we devise the gated heterogeneous fusion module (GHFM), which can automatically learn the varying contributions of the optical and SAR features for classifying different land types.

In this article, a novel heterogeneous modal fusion model called the collaborative attention-based heterogeneous gated fusion network (CHGFNet) is proposed, which consists of three main components: two-stream feature extractor, MCAM, and GHFM. The first component is divided into two parts; the first part is the pseudo-siamese network, which avoids the mutual interference between the network parameters of optical and SAR streams. The second part is the siamese network, which maps the deep features of the optical and SAR modalities into the shared space. The second and third components achieve the hierarchical fusion of both optical and SAR features to improve land cover classification. The overall architecture of CHGFNet is shown in Fig. 1.

We summarize our contributions as follows.

1) We propose a novel CHGFNet, which accomplishes the fusion of both optical and SAR features for land cover classification. Importantly, we intuitively study the complementary potential between the optical and SAR features based on the feature distribution histograms, which provides the reasonable support for the proposal of MCAM and GHFM.
2) To the best of our knowledge, this article first introduces MCAM and attempts to capture latent semantic interaction between optical and SAR features, which refers fine mutual attention maps for precisely focusing on the important areas of the optical and SAR feature maps.
3) We devise GHFM that automatically can learn different contributions of the optical and SAR features for different land categories, instead of simple score fusion with equal weights for the optical and SAR features.
4) We evaluate the proposed CHGFNet on three broad co-registered optical and SAR data sets, and achieve the competitive classification performance. Meanwhile, we also perform the detailed visualization to quantify two main components, the MCAM and GHFM.

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The rest of this article is organized as follows. Section II reviews related works, collaborative attention mechanism, and gated mechanism. We detail the proposed CHGFNet in Section III wherein each component is described. Section IV reports the complementary validation among optical and SAR features. Section V reports extensive comparative experiments on three broad co-registered optical and SAR data sets, and this article is concluded in Section VI.

II. Relate Work

Collaborative attention mechanism has been widely used in the natural language processing (NLP) [20] and the hashtag recommendation [22], [23], which can model the interactions between different modalities [17]. For example, Zhang *et al.* [22] proposed a collaborative attention network, which jointly performed the tweet-guided visual attention and image-guided textual attention to recommend hashtags for multimodal tweets. Li *et al.* [23] proposed topical collaborative attention networks for hashtag recommendation on microblogs, which modeled the content attention and topic attention, simultaneously. In the manner, the content representations were used to guide the topic attention, and the topic representation was used to guide the content attention. More recently, the collaborative attention mechanism has been expanded in vision tasks, such as person reidentification [18], [34] and video object segmentation [33]. Collaborative attention siamese network (COSNet) [33] is proposed, which incorporated a global collaborative attention mechanism to capture the scene context and global correlations. Wu *et al.* [18] proposed a deep collaborative attention-based comparators (DCCs), which consisted of the collaborative attention encoder for learning the co-dependent features. In this article, the MCAM is used to emphasize the importance of inherent correlation among optical and SAR features, which captures the complementary information between optical and SAR features to guide discriminative fusion features.

Gated Mechanism: In deep neural networks, especially the recurrent networks, long short-term memory (LSTM) networks [35], and gated recurrent unit (GRU) networks [36], the gated mechanism is commonly utilized to control message passing. Recently, the gated mechanism has been embedded into the multimodal deep learning networks to fuse different modalities for improving visual tasks [27], [28], [31], [37], [38]. Zeng *et al.* [30] proposed a novel model for multimodal learning based on the gated mechanism, which used the multiplicative gates to decide how modalities influence the unit activation for genre classification of movies. Zhang *et al.* [28] proposed a deep gated fusion network to generate a clear high-resolution frame from a single natural image with severe blur. Zhang *et al.* [31] proposed a novel adaptive collaborative attention network, which adopted the gated multimodal fusion module to fuse the textual and visual information for naming entity recognition in tweets. Locality-sensitive DeconvNet (LS-DeconvNet) was proposed, which introduced a novel gated fusion layer to effectively combine both RGB and depth features [37]. Three novel dynamic fusion methods were proposed to improve multimodal word representations, which utilized the modality-specific gate, category-specific gate, and sample-specific gate to, respectively, learn different weights of the linguistic and perceptual representations for each input modality, each super-sense category, and each word sample, respectively [38].

III. Methodology

In this article, we propose a novel multimodal fusion model called CHGFNet, which is used to learn the fusion features between optical and SAR images for realizing better land cover classification. The architecture of CHGFNet is shown in Fig. 1. We will describe our model in three parts: two-stream feature extractor, the MCAM, and GHFM.

A. Two-Stream Feature Extractor

Due to the advantages of DCNNs are hierarchical learning, automatic feature extraction, and weight sharing, DCNNs have been considered as most wide technique of machine learning in pattern recognition [39], [40]. A typical DCNN architecture includes three main components: convolution layer, pooling layer, and fully connected layer. Convolution layer divides the images into small blocks (receptive field) and extracts locally correlated pixel values by convolution kernels. Pooling layer sums up similar information in the neighborhood of receptive field, which generates invariant features and enlarges receptive field. Fully connected layer makes nonlinear combination of selected features, which are used for classification. In this article, we use two-stream CNN as the feature extractor to obtain deep semantic features of optical and SAR modalities, respectively.

Here, the limitation of sample number in the co-registered optical and SAR data sets, which makes it unrealistic to retrain an excellent network model. Fine-tuned models based on popular network architecture, such as AlexNet [41], VGG [42], and ResNet [43], have been used to address this limitation. Meanwhile, considering the proposed CHGFNet belongs to the patch-based land cover classification method, the optical and SAR patches are regarded as basic processing unit, whose sizes will determine the type of backbone network used in the CHGFNet. If the backbone network includes massive down-sampling layer (max-pooling layer), the downsampling factor will exceed the size of the input optical and SAR patches, which make features maps of last convolutional layer lose a lot of spatial information. Taking the PoDelta1 as an example, we can obtain the optimal result, when the size of optical and SAR patches is 29×29. Therefore, the VGG-16 and ResNet are not suitable as the backbone network in this work since their downsampling factor is 32. In this article, we select the AlexNet as backbone network.

Furthermore, two-stream feature extractor based on AlexNet is refined into two parts. In order to cope with the strongly different geometric and radiometric appearances of optical and SAR images, the first part adopts pseudo-siamese architecture without the fully connected layer for the shallow-level feature extraction. Comparing with the siamese network, the adopted pseudo-siamese network prevents the mutual interference between the optical and SAR streams [45], which is easily to

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learn the dominant features of the optical and SAR modalities, respectively. Then, the second part adopts the siamese network architecture to extract the high-level feature of optical and SAR modalities in the same share feature space.

The feature extraction process of the optical and SAR patches is as follows. Let X and Y represent the optical and corresponding SAR patches, respectively. For optical and SAR streams with $I = 5$ convolution layer, let $X^{(i)}$ and $Y^{(i)}$ denote the outputs of the ith layer in optical and SAR streams, respectively. $X^{(0)} = X$ and $Y^{(0)} = Y$ are the input samples; $X^{(5)}$ and $Y^{(5)}$ are the output feature maps of the 5th convolution layer. Each layer in optical and SAR streams computes the function $F_{\text{opt}}(\cdot)$ and $F_{\text{SAR}}(\cdot)$; the outputs $X^{(i)}$ and $Y^{(i)}$ of ith layer are computed as

$$X^{(i)} = F_{\text{opt}}\left(X^{(i-1)}; W_{\text{opt}}^{(i)}\right), \quad i = 1, \ldots, 5 \tag{1}$$

$$Y^{(i)} = F_{\text{SAR}}\left(Y^{(i-1)}; W_{\text{SAR}}^{(i)}\right), \quad i = 1, \ldots, 5 \tag{2}$$

where $W_{\text{opt}}^{(i)}$ and $W_{\text{SAR}}^{(i)}$ are the parameters of the ith layer in optical and SAR streams, respectively.

In order to reduce the internal covariance shift within the feature maps during the optimization, we, respectively, add a batch normalization (BN) [46] layer after the 5th layers of the optical and SAR streams. The normalized optical and SAR feature maps $\tilde{X} \in \mathbb{R}^{h\times w\times c}$ and $\tilde{Y} \in \mathbb{R}^{h\times w\times c}$ as follows:

$$\tilde{X} = \frac{X^{(5)} - E(X^{(5)})}{\sqrt{\delta_{\text{opt}} + \varepsilon}} \tag{3}$$

$$\tilde{Y} = \frac{Y^{(5)} - E\left(Y^{(5)}\right)}{\sqrt{\delta_{\text{SAR}} + \varepsilon}} \tag{4}$$

where δ_{opt} and δ_{SAR} depict the variations of $X^{(5)}$ and $Y^{(5)}$; $E(\cdot)$ is the mean operation; ε is the tiny amount; and h, w, and c are the height, width, and channel number of the feature maps, respectively.

B. MCAM

Based on the features provided in Section III-A, we have the optical feature $\tilde{X} \in \mathbb{R}^{h\times w\times \text{c}}$ and the corresponding SAR feature $\tilde{Y} \in \mathbb{R}^{h\times w\times c}$. Because both optical and SAR images are able to characterize the identity of the same areas, $\tilde{X}$ can provide the additional information that is not included in $\tilde{Y}$, which is helpful to improve the accuracies of land cover classification [6], [8], [9], [11], [13]. We adopt the MCAM to learn mutual attention between two modalities, which can capture the latent semantic interaction of different modalities. Fig. 2 shows the structure of MCAM. The details of this module are illustrated as follows.

First, to fuse two resource features codependently on each other, we compute the affinity matrix as follows:

$$M = \tilde{Y}^{\text{T}} W \tilde{X} \in \mathbb{R}^{(\text{hw})\times(\text{hw})} \tag{5}$$

where $W \in \mathbb{R}^{c\times c}$ is a trainable weight matrix. Here, $\tilde{X} \in \mathbb{R}^{(\text{hw})\times c}$ and $\tilde{Y} \in \mathbb{R}^{(\text{hw})\times c}$ are flattened into matrix representation. Each column $\tilde{X}_i$ in $\tilde{X}$ denotes the features vector at position $i \in \{1, \ldots, \text{hw}\}$ with c dimensions. Specifically, each element (i, j) in M represents the similarity between the features $\tilde{X}_i$ and $\tilde{Y}_j$. Then, we normalize M in row-wise to produce attention maps for the respective representation of $\tilde{X}$, as follows:

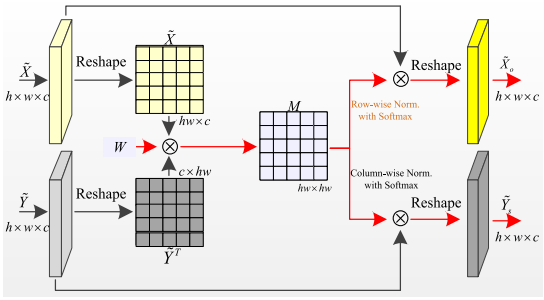


Fig. 2. Architecture of MCAM. $\otimes$ denotes the matrix multiplication.

$$M_{\text{o}} = \text{softmax}(M) \in \mathbb{R}^{(\text{hw})\times(\text{hw})} \tag{6}$$

where softmax$(\cdot)$ is the softmax function.

And we also normalize M in column-wise to produce attention maps for the respective representation of $\tilde{Y}$, as follows:

$$M_{\text{s}} = \text{softmax}(M^{\text{T}}) \in \mathbb{R}^{(\text{hw})\times(\text{hw})}. \tag{7}$$

With the attention maps, the attention summaries of optical features $\tilde{X}$ with respect to SAR features $\tilde{Y}$ can be computed as follows:

$$\widehat{X}_{\text{o}} = M_{\text{o}}\tilde{X} \in \mathbb{R}^{(\text{hw})\times c}. \tag{8}$$

In the same way, the attention summaries of SAR features $\tilde{Y}$ with respect to optical features $\tilde{X}$ can be computed as follows:

$$\widehat{Y}_{\text{s}} = M_{\text{s}}\tilde{Y} \in \mathbb{R}^{(\text{hw})\times c}. \tag{9}$$

In Section V-D1, we discuss MCAM to fuse co-dependent features for referring fine mutual attention maps, which is used to focus the important areas of the feature maps, and then obtain discriminative and robust features.

C. GHFM

Recently, the state-of-the-art fusion networks generally fuse both optical and SAR features with equal-weight score. In which, the feature concatenation and element-wise summation are frequently utilized schemes for multimodal feature fusion [24]. The problem of these basic fusion strategies is that the feature maps are fused together without measuring the varying contributions of different modalities in different land categories [24]–[26].

Therefore, the GHFM is proposed to effectively combine optical and SAR features. It consists of four layers, including a concatenation layer, a convolution layer, a fully connected layer with sigmoid function, and a feature summation layer, as shown in Fig. 3. Based on the features provided in Section III-B, we have the optical and SAR feature maps $\widehat{X}_{\text{o}}$ and $\widehat{Y}_{\text{s}}$, which are rewrote into 3-D tensor representation. After the feature concatenation, we obtain the fused map $\widehat{Z}$. Then, we employ a convolution layer with the $1 \times 1 \times c \times 2c$ convolution kernel for learning the correlations between

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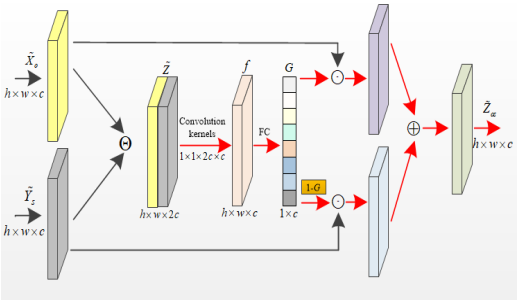


Fig. 3. Architecture of GHFM. FC denotes the fully connected layer.

two modalities, and obtain the fusion features f. Finally, the fully connected layer with sigmoid function is employed to obtain the gated vector G for optical features, and the gate vector $(1-G)$ for SAR feature s, which are used to automatically learn the varying contributions of different modalities and obtain the gated multimodal fusion features $\widehat{Z}_{\mathrm{os}}$. The equations for these operations are as follows:

$$\widehat{Z} = \widehat{X}_{\mathrm{o}} \Theta \widehat{Y}_{\mathrm{s}} \in \mathbb{R}^{h\times w\times 2c} \tag{10}$$

$$f = \mathrm{Conv}\left(\widehat{Z}\right) \in \mathbb{R}^{h\times w\times c} \tag{11}$$

$$G = \sigma\left(\widehat{w} \times \mathrm{Flat}(f) + \widehat{b}\right) \in \mathbb{R}^{1\times c} \tag{12}$$

$$\widehat{Z}_{\mathrm{os}} = G \odot \widehat{X}_{\mathrm{o}} \oplus (1-G) \odot \widehat{Y}_{\mathrm{s}} \in \mathbb{R}^{h\times w\times c} \tag{13}$$

where Θ denotes the feature concatenation, $\mathrm{Flat}(\cdot)$ is the flatten operation, $\widehat{b}$ is the bias, and $\sigma(\cdot)$ is the logistic sigmoid function. $\odot$ denotes the element-wise multiplication and $\oplus$ is the element-wise summation.

In Section V-D2, we will detail the proposed GHFM to automatically learn different contributions of optical and SAR features for classifying different land cover categories.

IV. Complementary Validation

Numerous fusion studies have shown that optical and SAR sources can provide the complementary information from individual sources, which is good for land cover classification [7], [8]. Especially, DCNNs have been used to the fusion of both optical and SAR features in RS object detection [47] and change detection [48], which have achieved the superior performance than traditional machine learning methods. However, DCNNs as the black box models, the deep semantic features extracted by DCNNs have worse interpretability than handcrafted features [49], [50], which makes it is difficult to design and select the optimal fusion methods. Furthermore, to the best of our knowledge, most previous fusion methods have demonstrated the complementary potential between different modalities based on the experimental accuracies, which did not provide intuitive complementary verification experiment.

Therefore, to intuitively study the complementary potential between the optical and SAR features $\tilde{X}$ and $\tilde{Y}$ extracted by (3) and (4), we compare the feature distribution histograms of different land categories within optical and SAR modalities in Figs. 4–6. Taking PoDelta1 data set (described in Section V-A) as an example, we randomly select 200 optical and SAR patch pairs of each land category as input samples to the proposed CHGFNet, and obtain the optical feature sets $\hat{X} = \{\hat{X}_i | i = 1, \ldots, K\}$ and the SAR feature sets $\hat{Y} = \{\hat{Y}_i | i = 1, \ldots, K\}$. In which, the optical and SAR feature sets of the ith land cover class are $\hat{X}_i = \{\tilde{X}_i^{(1)}, \ldots, \tilde{X}_i^{(200)}\}$ and $\hat{Y}_i = \{\tilde{Y}_i^{(1)}, \ldots . \tilde{Y}_i^{(200)}\}$; $\tilde{X}_i^{(j)}$ and $\tilde{Y}_i^{(j)}$ denote the optical and SAR features of the jth optical and SAR patch pair. Then, we use the principal component analysis (PCA) to extract the principal components $\check{X}_i \in \mathbb{R}^{128\times 200}$ and $\check{Y}_i \in \mathbb{R}^{128\times 200}$. Finally, we, respectively, show the distribution histograms of features $\check{X}$ and $\check{Y}$ for each land cover type in Fig. 4. Some phenomenon can be observed from the distribution: 1) the feature histograms of all land cover types can be observed with a nearly normal distribution; 2) the distributions of optical and SAR features for the same land type have partial overlapping; and 3) the optical and SAR features for all land types show a multimodal distribution with a clear separation. Especially, Fig. 4(c) demonstrates that optical and SAR features of water can reflect different attributes.

Meanwhile, we compare the features distribution histograms of different land cover types on the same modality in Figs. 5 and 6. Taking the water on PoDelta1 as an example, in the first row of Fig. 5, we compare the feature distribution histograms between water and other land types on optical modality; it is seen that the deep optical features of water and others are mostly overlapping, which are hard to be distinguished, except the water and tree in Fig. 5(c). Therefore, the optical features should not be used as dominant features for water classification. Furthermore, the feature distribution histograms of water and other land cover types on SAR modality are shown in the second row of Fig. 5; it is seen that the deep SAR features of water and others have a clear separation. We can find that SAR features of water and other land types are easily to be distinguished. Therefore, deep SAR features play an important role for water classification. Similarly, we also analyze the complementary between the optical and SAR features of the tree in Fig. 6. Some phenomenon is observed from feature distribution histograms, first, the feature distribution histograms of tree and other land types on optical modality, which have well distinguishable in the first row of Fig. 6. Meanwhile, from the second row of Fig. 6, it is seen that SAR features of water and others are overlapping, except the water and tree in Fig. 6(g). Therefore, the deep optical features are used as dominant features for the tree classification.

In a word, Fig. 4 shows the feature distribution histograms of the same land cover types on the optical and SAR modalities which are bimodal. Figs. 5 and 6 have confirmed that the optical and SAR features of the water and tree have the complementary potential. The complementarity verification of other land types is similar to the water and tree, which also provides the reasonable explanation for the proposed MCAM and GHFM.

V. Experiments and Discussion

In this section, the proposed CHGFNet is evaluated on three broad co-registered optical and SAR data sets.

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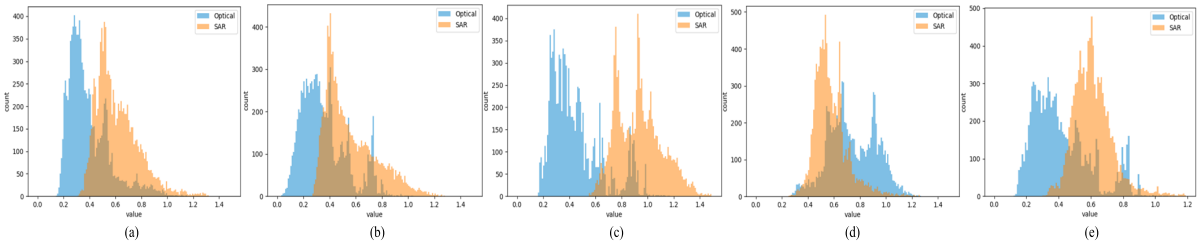


Fig. 4. Statistical distributions of the optical and corresponding SAR features for different land cover types on PoDelta1. (a) Statistical distributions of the building features. (b) Statistical distributions of the farmland features. (c) Statistical distributions of the water features. (d) Statistical distributions of the tree features. (e) Statistical distributions of the unknown features.

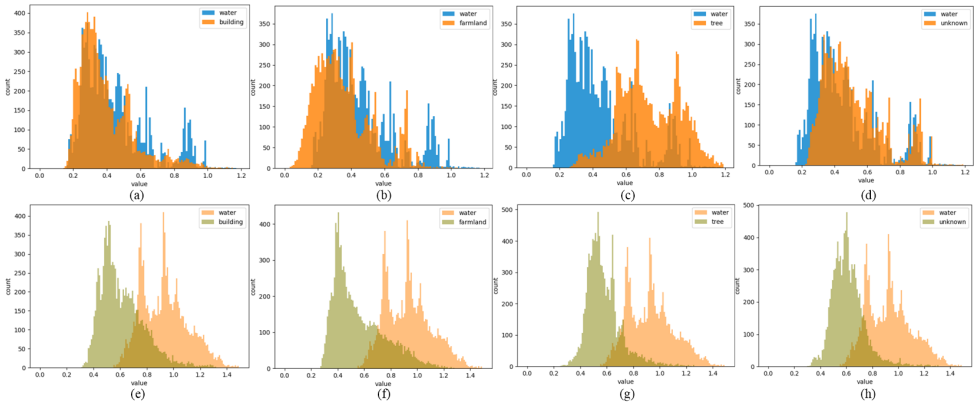


Fig. 5. Feature statistical distributions of water and other land cover types on PoDelta1. (a)-(d) Compare the feature statistical distribution between the water and other land cover types (building farmland, tree and unknown) on optical modality, respectively. (e)-(h) Compare the feature statistical distribution between the water and other land cover types (building farmland, tree and unknown) on SAR modality, respectively. We find that the deep optical features of water and other land cover types are mostly overlapping and hard to distinguish; however, the SAR features of water and others are clearly distinguishable. Therefore, the optical and SAR patches have the complementary potential in water classification.

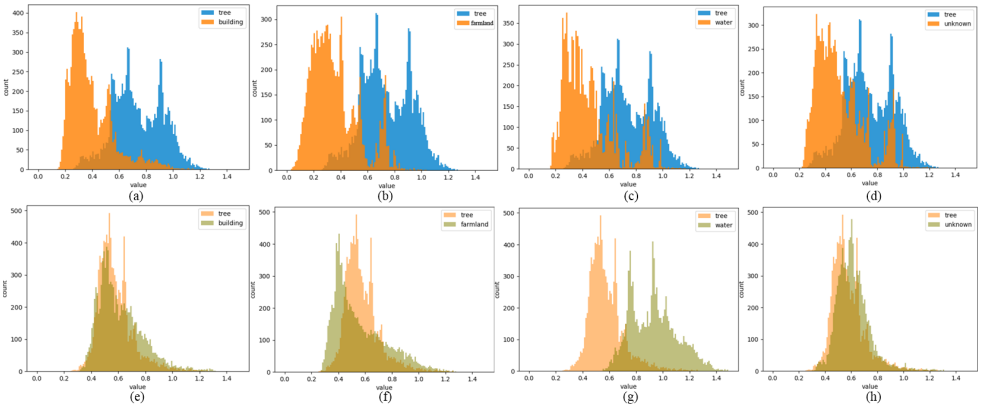


Fig. 6. Feature statistical distributions for tree and other land cover types on PoDelta1. (a)-(d) Compare the feature statistical distribution between the water and other land cover types (building farmland, water and unknown) on optical modality, respectively. (e)-(h) Compare the feature statistical distribution between the water and other land cover types (building farmland, water and unknown) on SAR modality, respectively. We find that the SAR features of tree and other land types are mostly overlapping and hard to distinguish; however, the optical features of tree and other are clearly distinguishable. Therefore, the optical and SAR patches have the complementary potential in tree classification.

First, the experimental setup is described in Section V-A, which includes the description of three broad data sets, the implementation details, and the evaluation metrics of land cover classification. In Section V-B, we provide the description of baseline methods. The configurations of the significant hyper-parameters are described in Section V-C. In Section V-D, the ablation experiments are described to verify the validity of the MCAM and GHFM, respectively. In Section V-E, we compare land cover classification results of the CHGFNet and baseline methods on three broad co-registered optical and SAR data sets. Finally, we compare the qualitative visualizations of the features extracted by the different methods in Section V-F, which can intuitively explain the superiority of the CHGFNet.

A. Experimental Setup

1) Data Sets: We demonstrate the effectiveness of our model with extensive experiments on three broad co-registered

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TABLE I
LAND TYPES DISTRIBUTION ON THE PoDelta1

Class No.	Land Types	Number of Samples	Percentage
1	Building	213868	2.99%
2	Farmland	3834510	53.58%
3	Water	826053	11.55%
4	Tree	1784183	24.93%
5	Unknown	497418	6.95%

TABLE II
LAND TYPES DISTRIBUTION ON THE PoDelta2

Class No.	Land Types	Number of Samples	Percentage
1	Building	196268	0.85%
2	Farmland	9020149	39.24%
3	Water	10058772	43.75%
4	Tree	299079	1.30%
5	Unknown	3415161	14.86%

optical and SAR data sets. The first two co-registered optical and SAR data sets are two different regions of PoDelta, Italy, which are termed the PoDelta1 and PoDelta2. The sizes of PoDelta1 and PoDelta2 are 3666 × 1952 and 4491 × 5119, respectively. And their resolution is 2.5 m × 2.5 m. The SAR images with an HH-polar imaging mode are the COSMO-SkyMed geocoded ground range (GEC) level 1C products and in X-band, while the corresponding optical images are obtained from Google Earth, whose spatial resolution is slightly lower than the SAR image. The PoDelta1 and PoDelta2 data sets have been used to land cover classification in [51], which has confirmed the credibility of the PoDelta1 and PoDelta2 data sets in land cover classification. Moreover, to evaluate the CHGFNet, we relabel the ground truths of the PoDelta1 and PoDelta2 provided by Ren *et al.* [51] based on the Open Street Maps [52], Urban Atlas [53], and the visual interpretation, and create finer ground truths. All the pixels of PoDelta1 and PoDelta2 are assigned to five categories: building, farmland water, tree, and unknown.

The third co-registered optical and SAR data set is the area of Wuhan, Hubei Province, China, which is termed WuHan. It has 1039 × 886 pixels in size, and the ground resolution is 3.0 m × 3.0 m. In which, the SAR image with an HH-polar imaging mode is geometrically rectified multispectral Level-1C reflectance products and in X-band, and it comes from the TerraSAR. The corresponding optical image is obtained from the Google Earth. The ground truth map of WuHan data set is drawn according to the field research that were carried out to collect ground truth and phenology information of the concerned fields, and the credibility of ground truth map has confirmed in [4]. All the pixels are assigned to eight categories: resident (Re), industrial land (Ind), water (W), tree (T), farmland (Fa), background (Bg), unhardened road (Unr), and asphalt road (As). And Tables I–III show land-type distributions on the PoDelta1, PoDelta2, and WuHan data sets; it is seen that three broad co-registered optical and SAR data sets have the problem of class imbalance.

For the co-registration of optical and SAR images on above three data sets, we adopt the coarse-to-fine registration strategy

TABLE III
LAND TYPES DISTRIBUTION ON THE WuHan

Class No.	Land Types	Number of Samples	Percentage
1	Resident	63730	6.93%
2	Farmland	158980	17.27%
3	Water	154211	16.75%
4	Tree	178188	19.36%
5	Unhardened road	34174	3.71%
6	Industrial land	118496	12.87%
7	Asphalt road	21573	2.34%
8	Background	191202	20.77%

TABLE IV
PA, MPA, MIOU, AND FWIOU OF DIFFERENT METHODS ON PoDelta1

Methods	PA(%)	MPA(%)	MIOU(%)	FWIoU(%)
CNN-Opt.	90.86	87.11	81.09	84.43
CNN-SAR	80.17	80.14	61.23	68.96
CNN-MCAM-Con.	93.01	90.65	84.78	87.86
CNN-MCAM-Pro.	93.79	91.31	85.15	89.26
CNN-CBAM-GHFM	96.01	96.12	87.16	92.03
2D-CNN[9]	91.94	88.42	82.15	85.99
MRSDC[56]	94.89	95.94	85.38	91.31
CHGFNet	**97.86**	**97.11**	**88.59**	**93.54**

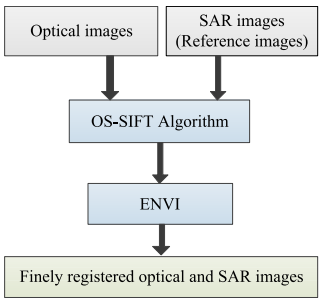


Fig. 7. Co-registration flowchart of the optical and SAR images.

to complete fine registration. Fig. 7 shows the co-registration flowchart of the optical and SAR images. First, to better the co-registration of the optical and SAR images, we download optical images of level 17 from Google Earth, whose spatial resolution 2.15 m × 2.15 m is slightly lower than SAR image. Then, taking SAR image as the reference image, the optical-to-SAR SIFT-like (OS-SIFT) algorithm [54] is used to complete the coarse registration of the optical and SAR images. Finally, lots of corresponding points between the optical and SAR images are selected manually to achieve fine registration based on the ENVI software.

2) Implementation Details: The entire network is trained by using stochastic gradient descent (SGD) optimizer, the initial learning rate is set to 1.0×10^{-4} for the PoDelta1 and PoDelta1, and the initial learning rate is set to 1.0×10^{-5} for the WuHan. During the training process, the batch size is set to 32. In order to improve the training capability of our model, we use an exponentially decaying learning rate with a decay rate of 0.95 in later epochs, which is expressed as: $l = 0.0001 \times (0.95)^{m/N}$, where m is the total number of mini-batches that have been processed for training and N is the number of min-batches in one epoch, and the epoch is

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50. We implement the whole algorithm with Tensorflow. All experiments are conducted on the platform of Window 10 with Intel Corn i9-9900KF 3.60-GHz CPU, NVIDIA GeForce RTX 2080Ti GPU, and 16-GB memory.

3) Evaluation Metrics: We report four metrics from common semantic segmentation and scene parsing evaluations, including the pixel accuracy (PA), the mean pixel accuracy (MPA), the mean intersection over union (MIoU), and the frequency weighted intersection over union (FWIoU). Let n_{ij} be the number of pixels of class i predicted to belong to class j, n_{ii} is the number of pixels of class i predicted to belong to class i, $t_i = \sum_j^N n_{ij}$ is the total number of pixels of class i, and there are K different land type classes. The PA, MPA, MIoU, and FWIoU are calculated as follows:

$$\text{PA} = \frac{\sum_i^K n_{ii}}{\sum_i^K t_i} \tag{14}$$

$$\text{MPA} = \frac{1}{K}\sum_i^K \frac{n_{ii}}{t_i} \tag{15}$$

$$\text{MIoU} = \frac{1}{K}\sum_i^K \frac{n_{ii}}{t_i + n_{ji} - n_{ii}} \tag{16}$$

$$\text{FWIoU} = \frac{1}{\sum_{k=1}^K t_k}\sum_i^K \frac{t_i n_{ii}}{t_i + n_{ji} - n_{ii}} \tag{17}$$

where the PA, MPA, MIoU, and FWIoU can describe the global classification performance. And the MPA, MIoU, and FWIoU are able to avoid land cover classification bias, due to the classes' imbalance on PoDelta1, PoDelta2, and WuHan data sets, which provide more credible assessment for the CHGFNet.

B. Baseline Methods

In this section, we will describe the proposed CHGFNet in the comparisons of our main experiments. Our experiment mainly concerns three groups of comparison models: the unimodal deep learning models, the variant models of our method, and previous state-of-the-art models. In above models, we use the AlexNet [41] pretrained on ImageNet [44] as the backbone network. The models are listed as follows.

1) Unimodal Deep Learning Models: To illustrate our model can automatically learn the varying contributions of the optical and SAR modalities to efficiently handle multimodal features fusion, we compare our model with the following unimodal deep learning models, which only explore the impact of single source features for land cover classification.

1) *CNN+Optical*: We replace the last fully connected layer of AlexNet with randomly initialized K-way classification layer. The optical image patches are the input samples. We refer this method as "CNN-Opt."
2) *CNN+SAR*: Similar to the network structure of "CNN-Opt.," but in this method, the SAR patches are the input samples. We refer this method as "CNN-SAR."

2) Variant Models: To analyze the contribution of each component in our model, we ablate the full model and demonstrate the effectiveness of each component.

1) *CNN+MCAM+Concatenation:* This model is a variant of our model without the GHFM. It directly concatenates the features from different modalities to obtain the fusion features for land cover classification. We refer this model as "CNN-MCAM-Con."
2) *CNN+MCAM+Element-Wise Product:* This model is a part of our model without the GHFM. Instead, we use the element-wise production to combine both optical and SAR features. We refer this model as "CNN-MCAM-Pro."
3) *CNN+CBAM+GHFM:* This model is also a variant of our model without the MCAM. Instead, the channel-wise attention modules of the CBAM [55] is embedded behind each stream, respectively. We refer this method as "CNN-CBAM-GHFM."

3) State-of-the-Art Models: To illustrate how well our model can handle the land cover classification task based on optical and SAR data, we compare our model with the following existing state-of-the-art models.

1) *2-D-CNN:* A multimodal deep network was proposed, which was used to fuse both optical and SAR images for the high-quality performance crop type classification [9]. To adapt this model to our multisource scenario, we connect the optical and SAR images to obtain the four-channel inputs. Then, we add one convolution layer before the AlexNet pretrained on ImageNet. Finally, we replace the last fully connected layer of AlexNet with a randomly initialized K-way classification layer for land cover classification. Also, in this case, this competitor is learned with end-to-end. We refer this method as "2-D-CNN."
2) *Multisource remote sensing data classification (MRSDC):* Xu *et al.* [56] proposed two-brand convolution network for multisource RS data classification. In which, a two-tunnel convolution network was developed to extract spectral–spatial features from hyperspectral imagery (HIS) data; the CNN with cascade block was designed for feature extraction from LiDAR data. To adapt such model to our multisource scenario, we use a two-tunnel convolution network to deal with optical images, and the CNN with cascade block is designed for the feature extraction from SAR images. We refer this method as "MRSDC."

C. Configuration of Hyperparameters

We determine the following two crucial hyperparameters by the experiment's performance: the number of training samples and the size of training samples. The specific analysis of them on the PoDelta1 data set will be elaborated in Section III-C. The determining process of the two crucial hyperparameters of the PoDelta2 and WuHan data sets is the same as the experiment on the PoDelta1.

1) Number of Training Samples: In three broad co-registered optical and SAR data sets, the amount of land cover categories is unbalanced; it is reasonable to adopt the selecting strategy that identical samples amount of each land category is chosen in our experiments, which

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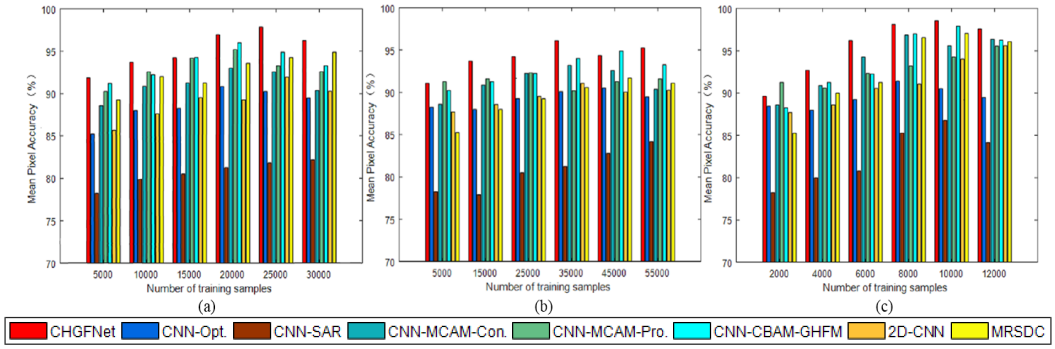


Fig. 8. Histograms to depict the trend of classification accuracies with the amount of training samples on three data sets. (a) Classification accuracies are given based on different numbers of training samples on PoDelta1. (b) Classification accuracies are given based on different numbers of training samples on PoDelta2. (c) Classification accuracies are given based on different numbers of training samples on WuHan.

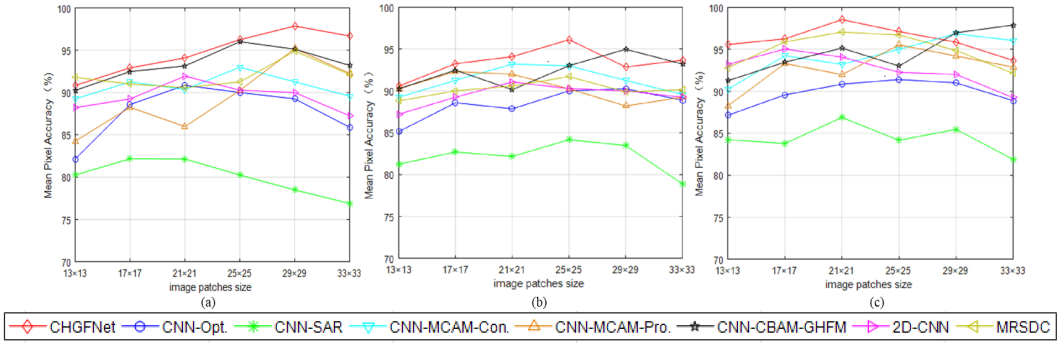


Fig. 9. Curves to depict the trend of classification accuracies with the size of training samples on three data sets. (a) Accuracy curves are given based on different sizes of training samples on PoDelta1. (b) Accuracy curves are given based on different sizes of training samples on PoDelta2. (c) Accuracy curves are given based on the size of training samples on WuHan.

can ensure the unbiased training of the network. First, the 29 × 29 samples are randomly picked for every land category in the optical and SRA patch pairs. The trend of the classification accuracies with the amount of samples is presented in Fig. 8(a), the horizontal axis {5000, 10 000, 15 000, 20 000, 25 000, 30 000} denotes the number of training samples, and the vertical axis denotes the MPA. It is easy to find that the accuracy of our CHGFNet increases with the samples number from 5000 to 20 000. While the classification accuracies are almost unchanged when the number of training samples is from 20 000 to 25 000, which indicates that the 20 000 optical and SAR patches already contain various patterns and structures in our predefined five land cover categories of PoDelta1. And when the number of patches is 25 000, the CHGFNet has optimal land cover classification accuracy.

Similarly, Fig. 8(b) and (c) shows the determination of the number of training samples on the PoDelta2 and WuHan data sets. It is seen that we can achieve the highest accuracies for our CHGFNet, when the number of training samples on the PoDelta2 and WuHan data sets is 35 000 and 10 000, respectively.

2) Size of Training Samples: In our experiments, every patch pair is regarded as a basic processing unit, whose size determines whether the spatial information included in optical and SAR patches is good for extracting discriminative fusion features in the proposed model. The size of samples is determined by the classification accuracy in Fig. 9, the horizontal axis denotes the image patches size, its value ranges from 13 × 13 to 33 × 33 with a span of 4, and the vertical axis denotes the MPA. It can be viewed that the accuracy of our CHGFNet increases with the size of samples from 13 × 13 to 29 × 29. While the size of samples is 29 × 29, the classification accuracy is highest, and the accuracy decreases with the size of samples from 29 × 29 to 33 × 33, which indicates that 29 × 29 patch contains the adequate and appropriate spatial information for predefined five land categories of PoDelta1.

Meanwhile, Fig. 9(b) and (c) shows the optimal the size of training sample for the PoDelta2 and WuHan data sets. It is easy to find that we can achieve the highest accuracies, when the sizes of training samples on the PoDelta2 and WuHan data sets are 25 × 25 and 21 × 21, respectively.

D. Components Validation

1) Validation of MCAM: To clearly investigate the MCAM can capture the sufficient interactions between optical and SAR features to refer fine attention maps, the attention maps are used to precisely focus on the important areas of the feature maps $\widehat{X}_{\mathrm{o}}$ and $\widehat{Y}_{\mathrm{s}}$; we compare the MCAM with channel-wise attention mechanism of CNN-convolutional block attention module (CABM)-GHFM by visualizing the randomly selected 25 channels of the feature $\widehat{X}_{\mathrm{o}}$ and $\widehat{Y}_{\mathrm{s}}$ in Figs. 10–12.

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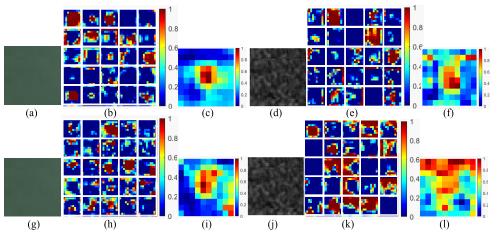


Fig. 10. Visualization of the randomly selected 25 channels of water features $\widehat{X}_{\mathrm{o}}$ and $\widehat{Y}_{\mathrm{s}}$ extracted by CABM and MCAM on Podelta1, respectively. (Top) Visualization results for CABM. (Bottom) Visualization results for MCAM. (a) and (g) Same optical patches of water. (d) and (j) Same SAR patches of water. (b) and (h) Visualization results of randomly selected 25 channels in the feature maps $\widehat{X}_{\mathrm{o}}$ extracted by CABM and MCAM, respectively. (e) and (k) Visualization of randomly selected 25 channels in the feature maps $\widehat{Y}_{\mathrm{s}}$ extracted by CABM and MCAM, respectively. (c), (i), (f), and (l) are obtained based on (b), (e), (h), and (k).

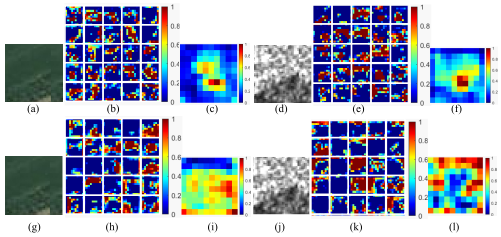


Fig. 11. Visualization of the randomly selected 25 channels of farmland features $\widehat{X}_{\mathrm{o}}$ and $\widehat{Y}_{\mathrm{s}}$ extracted by CABM and MCAM on Podelta1, respectively. (Top) Visualization results for CABM. (Bottom) Visualization results for MCAM. (a) and (g) Same optical patches of farmland. (d) and (j) Same SAR patches of farmland. (b) and (h) Visualization results of randomly selected 25 channels in the feature maps $\widehat{X}_{\mathrm{o}}$ extracted by CABM and MCAM, respectively. (e) and (k) Visualization of randomly selected 25 channels in the feature maps $\widehat{Y}_{\mathrm{s}}$ extracted by CABM and MCAM, respectively. (c), (i), (f), and (l) are obtained based on (b), (e), (h), and (k).

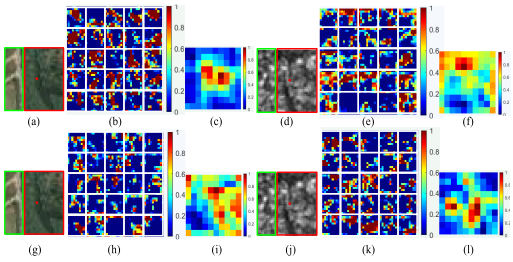


Fig. 12. Visualization of the randomly selected 25 channels of farmland features $\widehat{X}_{\mathrm{o}}$ and $\widehat{Y}_{\mathrm{s}}$ extracted by CABM and MCAM on Podelta2, respectively. (Top) Visualization results for CABM. (Bottom) Visualization results for MCAM. (a) and (g) Same optical patches of farmland. (d) and (j) Same SAR patches of farmland. (b) and (h) Visualization results of randomly selected 25 channels in the feature maps $\widehat{X}_{\mathrm{o}}$ extracted by CABM and MCAM, respectively. (e) and (k) Visualization of randomly selected 25 channels in the feature maps $\widehat{Y}_{\mathrm{s}}$ extracted by CABM and MCAM, respectively. (c), (i), (f), and (l) are obtained based on (b), (e), (h), and (k).

We discuss the superiority of the MCAM in detail, according to the input samples containing one land cover category and the input samples containing multiple land cover categories, respectively.

For input samples containing a single land cover category, $\widehat{X}_{\mathrm{o}}$ and $\widehat{Y}_{\mathrm{s}}$ should be fully activated to obtain global and robust features. Figs. 10 and 11 visualize the randomly selected 25 channels in feature maps $\widehat{X}_{\mathrm{o}}$ and $\widehat{Y}_{\mathrm{s}}$ extracted by the MCAM and the channel attention mechanism of CNN-CABM-GHFM, respectively. Taking Fig. 10 as an example, we can know that the first row denotes the features visualization from the channel attention of CNN-CABM-GHFM, and the second row denotes the features visualization from MCAM. The first three column denotes the optical features visualization from the above two attention mechanisms, and the last three column denotes the SAR features visualization from the above two attention mechanisms. In which, Fig. 10(a) and (g) shows the same optical patches of the water, and Fig. 10(d) and (i) shows corresponding SAR patches of the water. Fig. 10(b) shows the visualization of the randomly selected 25 channels in feature maps $\widehat{X}_{\mathrm{o}}$ from the channel attention of CNN-CABM-GHFM, and Fig. 10(h) shows the visualization of the randomly selected 25 channels in feature maps $\widehat{X}_{\mathrm{o}}$ from the MCAM. From Fig. 10(b) and (h), we can find that the 25 channels have more activated areas in Fig. 10(h), which means the MCAM can capture the inherent correlation among optical and SAR features to fully highlight the interest area of feature maps. Similarly, Fig. 10(e) shows the visualization of the randomly selected 25 channels in the feature maps $\widehat{Y}_{\mathrm{s}}$ from the channel attention of CNN-CABM-GHFM, and Fig. 10(k) shows the visualization of the randomly selected 25 channels in the feature maps $\widehat{Y}_{\mathrm{s}}$ from MCAM; we also can find that 25 channels have more activated areas in Fig. 10(k). Meanwhile, to better understand the contributions of the MCAM, we perform the following operations on Fig. 10(b), (e), (h), and (k) to obtain Fig. 10(c), (f), (i), and (l), respectively:

$$P_{\mathrm{s}} = \sum_{i=1}^{25} q_i \tag{18}$$

$$\tilde{P}_{\mathrm{s}} = \frac{P_{\mathrm{s}} - \min(P_{\mathrm{s}})}{\max(P_{\mathrm{s}}) - \min(P_{\mathrm{s}})} \tag{19}$$

where q_i is the ith channel of randomly selected 25 channels, min(.) denotes the minimum operation, and max(.) denotes the maximum operation.

Furthermore, we can clearly find that the activated areas in Fig. 10(i) are obviously more than the activated areas in Fig. 10(c). Similarly, Fig. 10(l) has more activated areas than activated areas of in Fig. 10(f). The analysis for Fig. 11 is similar to Fig. 10. From Fig. 11, we can also obtain the similar conclusions as Fig. 10. Therefore, MCAM can precisely focus on the important areas of feature maps to guide the learning of robust information.

For input samples containing multiple land cover categories, the target areas in $\widehat{X}_{\mathrm{o}}$ and $\widehat{Y}_{\mathrm{s}}$ should be activated, and the nontarget area should be suppressed. The meanings of each subpicture in Fig. 12 is the same as each subpicture in Figs. 10 and 11. The green box denotes the farmland, and the red box denotes the unknown classes from the PoDelta2 in Fig. 12(a), (d), (g), and (j). And the input sample is classified into the unknow class, namely, the label of the red point (center point) is classified as the unknow class. Therefore, the left areas of Fig. 12(c), (f), (i), and (l) corresponding to

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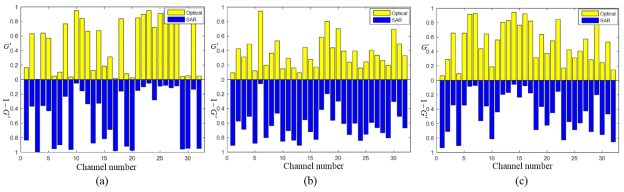


Fig. 13. Visualization of the mean gated vectors G' and the vectors $(1 - G')$ for building, water, and tree of the PoDelta1. (a) Visualization for the building. (b) Visualization for the water. (c) Visualization for the tree.

the green box should be in inhibition, while the right areas corresponding to the red box should be activated. As shown, the left areas of Fig. 12(i) have a small number of activated units, but most activated units are located in the right areas of Fig. 12(i) and are more than the right areas of Fig. 12(c), which means the MCAM can activate the target area of $\widehat{X}_{\mathrm{o}}$ more comprehensively. Meanwhile, it is seen that most activated units are distributed in the right areas of Fig. 12(l). However, the left areas of Fig. 12(f) have lots of active elements, which denotes the channel attention of the CNN-CABM-GHFM cannot stand out the discriminative areas of feature maps very well in the classification for input samples containing multiple land cover categories.

In summary, it is clear that the MCAM actually utilizes the collaborative attention mechanism to realize the sufficient interactions between optical and SAR features. For input samples containing a single land category, MCAM can fully activate the feature maps that contain the globe and robust information. Meanwhile, for input samples containing multiple land categories, the MCAM can selectively highlight the target areas of features maps.

2) Validation of GHFM: To analyze the GHFM can automatically learn the varying contributions of the optical and SAR features for classifying different land categories, we visualize the mean gated vectors G' and the vectors $(1 - G')$ for different land types on PoDelta1 and WuHan data sets in Figs. 13 and 14. First, we randomly select 200 optical and SAR patch pairs as the input samples to the proposed CHGFNet, and obtain gated vectors $G \in \mathbb{R}^{256\times 200}$. Furthermore, we take the average of G in the column as follows:

$$\bar{G} = E(G) \tag{20}$$

where $E(\cdot)$ is the mean operation.

Then, we randomly select 32 values from $\bar{G}$ to obtain the mean gated vectors G' for optical features and the vectors $\left(1 - G'\right)$ for SAR features. Finally, we visualize them in Figs. 13 and 14. The abscissa axis denotes the number of selected channels, and the horizontal axis is divided into upper and lower parts, the upper part denotes the mean gated vectors G', and the lower part denotes the vectors $\left(1 - G'\right)$.

In Fig. 13, taking the building, water, and tree as examples, we can find that different channels in each modality feature map have different weights. And the GHFM can achieve that the corresponding channels in optical and SAR feature maps have different contributions for classifying different land categories. Especially, for the water, most SAR channels have greater weights than the optical channels, which denotes the SAR sources are dominant and have higher contributions for classifying water in Fig. 13(b). For the tree, most optical channels have greater weights than the SAR channels, which denote the optical source plays more an important role for the classification of the tree in Fig. 13(c). And from the description of Figs. 5 and 6 in Section IV, we can also obtain the same conclusion, which confirms the GHFM automatically learns different contributions for the optical and SAR features. Similarly, to further test the abilities of the GHFM, we use a more challenging co-registered optical and SAR data set called WuHan data set. In Fig. 14, we show the inspections on six land cover types of WuHan data set, the resident, industrial land, tree, farmland, background, and asphalt road. From Fig. 14, we also find the similar conclusions to Fig. 13.

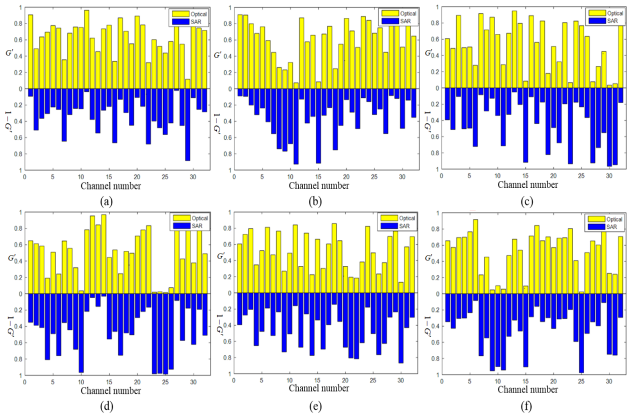


Fig. 14. Visualization of the mean gated vectors G' and the vectors $(1 - G')$ for different land types of the WuHan data set. (a) Visualization for the resident. (b) Visualization for the industrial land. (c) Visualization for the tree. (d) Visualization for the farmland. (e) Visualization for the background. (f) Visualization for the asphalt road.

In summary, it is clear that GHFM actually utilizes the gated mechanism to pass important information of optical and SAR modalities. The GHFM not only can automatically highlight the important channel of single modal features, but also learn the varying contributions of the optical and SAR modalities for classifying different land categories. Meanwhile, we observe that the optical modal has higher contributions than SAR modal for the classification of most land types on PoDelta1 and WuHan data sets.

E. Classification on Three Image Data Sets

To verify the effectiveness of the proposed CHGFNet and the discrimination of the learned fusion features, two unimodal deep learning models, three variant models, and two start-of-the-art models are adopted as contrast methods, which are described in Section V-B. The input samples of all contrast methods are raw image data, which are extracted into different features by these different methods. Then, the features are fed to the softmax function for the land cover classification. The classification accuracies of test samples are shown in Tables IV–VI. And the classification maps are shown in Figs. 15–17.

1) Classification Results on PoDelta1: In the experiment of PoDelta1, two crucial hyperparameters are given: the size of

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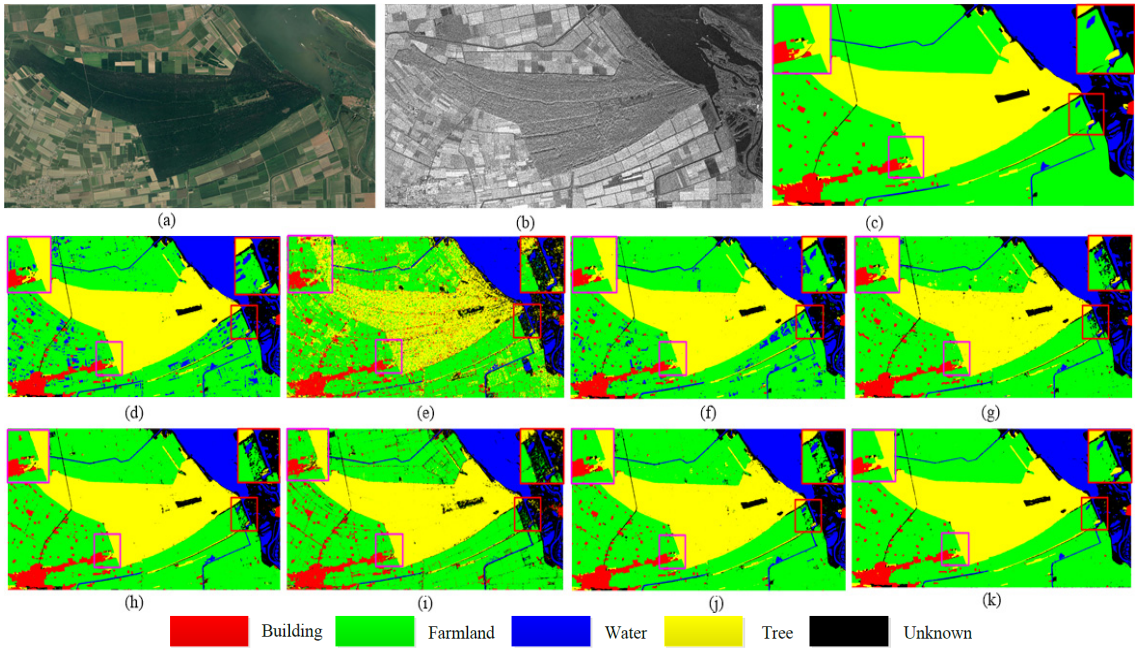


Fig. 15. Classification maps of different methods on PoDelta1. (a) Original optical image. (b) Original SAR image. (c) Ground truth. (d) CNN-Opt. (e) CNN-SAR. (f) CNN-MCAM-Con. (g) CNN-MCAM-Pro. (h) CNN-CBAM-GHFM. (i) 2-D-CNN. (j) MRSDC. (k) CHGFNet.

TABLE V
PA, MPA, MIoU, AND FWIoU OF DIFFERENT METHODS ON PoDelta2

Methods	PA(%)	MPA(%)	MIOU(%)	FWIoU(%)
CNN-Opt.	90.25	93.05	70.78	84.84
CNN-SAR	84.17	88.43	66.76	75.29
CNN-MCAM-Con.	93.21	94.12	74.78	87.54
CNN-MCAM-Pro.	92.32	93.14	73.54	86.25
CNN-CBAM-GHFM	94.98	95.42	80.67	90.97
2D-CNN[9]	91.07	93.09	67.45	83.83
MRSDC[56]	92.71	93.96	73.85	86.86
CHGFNet	**96.12**	**96.46**	**83.83**	**92.81**

TABLE VI
PA, MPA, MIoU, AND FWIoU OF DIFFERENT METHODS ON WuHan

Methods	PA(%)	MPA(%)	MIOU(%)	FWIoU(%)
CNN-Opt.	91.41	89.89	83.39	84.57
CNN-SAR	86.89	83.90	74.52	74.80
CNN-MCAM-Con.	96.86	96.67	93.76	93.94
CNN-MCAM-Pro.	95.54	94.31	90.33	91.49
CNN-CBAM-GHFM	97.90	97.36	95.25	95.92
2D-CNN[9]	95.60	95.31	91.00	91.70
MRSDC[56]	97.06	96.69	93.82	94.35
CHGFNet	**98.69**	**98.54**	**96.87**	**97.35**

sample patches is 29 × 29 and the training sample number of each land cover type is 25 000. The detailed description is shown on the aforementioned experiments in Section V-C. In order to objectively evaluate the performance of our method, four quantitative metrics of our model and comparison methods are shown in Table IV. Meanwhile, the classification maps are shown in Fig. 15.

Table IV shows four quantitative metrics of our model and comparison methods, and we have gathered several experiment findings. First, it is obvious that the proposed CHGFNet trained on PoDelta1 achieves a better performance compared with unimodal deep learning models such as the CNN-SAR and CNN-Opt., with 17.0% improvement, which verifies the CHGFNet can fully capture the complementary information between optical and SAR modalities for improving the land cover classification. Second, all the components of our model play an important role for improving performance. If any component is missing, then the performance will decrease. CNN-MCAM-Con. is a variant of our model without the GHFM instead of the feature connection fusion. Compared with our model's performance, the PA is reduced by 4.85%. CNN-MCAM-Pro. is also a variant of our model without the GHFM instead of the feature multiplication fusion. The PA of this model is decreased by 4.08%. Our model improves a lot by adaptively considering the varying contributions of the two modalities on delineating different categories in different scenes based on the GHFM, which includes the independent optical channel attention and SAR channel attention, so that the semantic interaction between different modalities cannot be captured. The PA is dropped by 1.35% compared with our model. Therefore, the multimodal collaborative attention is used to realize the interaction of optical and SAR modalities, which is necessary for our model. Third, by comparing the state-of-the-art models such as the 2-D-CNN and MRSDC, it is obvious that our CHGFNet achieves better performance, with 2.97% improvement. In a word, the CHGFNet consists of three components: the feature extractor based on the pseudo-siamese and siamese neural networks, the MCAM, and the GHFM. The two-stream feature extractor avoids that network parameters are tempted to ingratiate dominant modal data and integrate optical and SAR features. Specifically, the MCAM can learn mutual attention between the optical and

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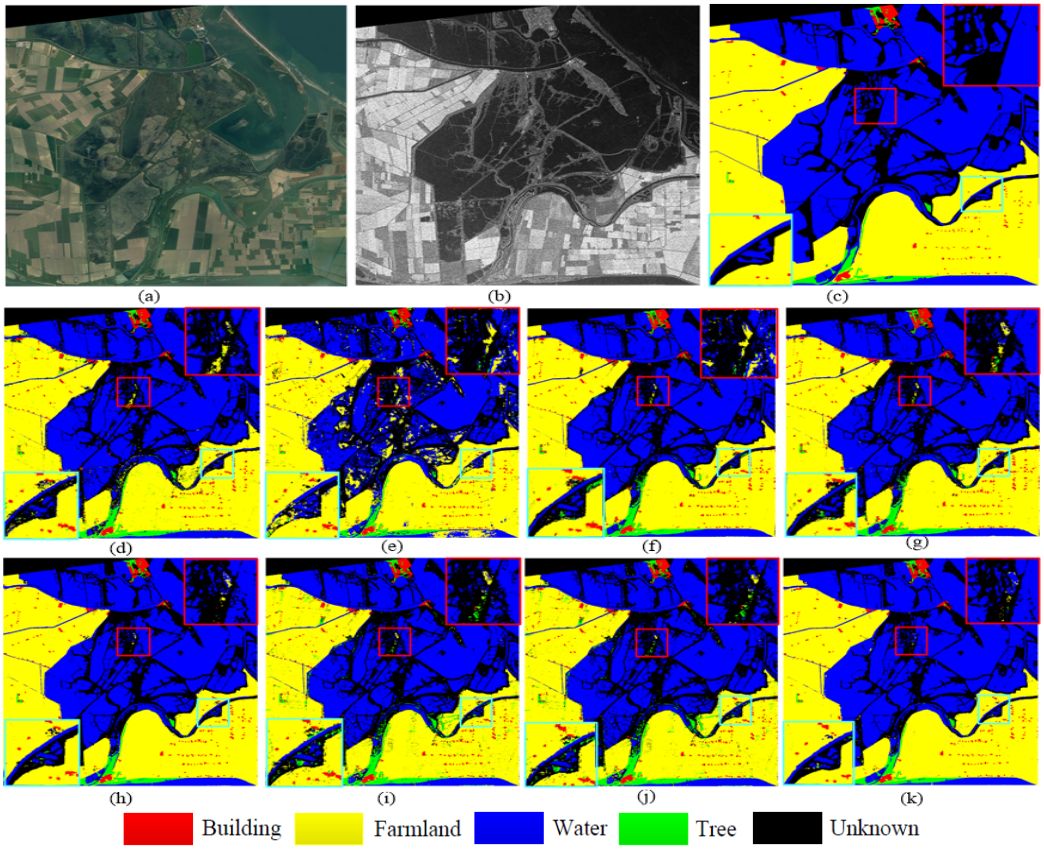


Fig. 16. Classification maps of different methods on PoDelta2. (a) Original optical image. (b) Original SAR image. (c) Ground truth. (d) CNN-Opt. (e) CNN-SAR. (f) CNN-MCAM-Con. (g) CNN-MCAM-Pro. (h) CNN-CBAM-GHFM. (i) 2-D-CNN. (j) MRSDC. (k) CHGFNet.

SAR features, and the GHFM can explore varying contributions of optical and SAR features for different land categories to realize feature fusion.

Meanwhile, Fig. 15 shows the classification maps. We reveal that the classification map derived from our method is better than other contrast methods. In order to fully compare these land cover classification maps, we enlarge some details with the red and rose regions in Fig. 15. In red region, it can be seen that the farmland is misclassified into unknown area in the contrast methods. In contrast, the proposed CHGFNet produces fewer misclassification. In the rose region, the farmland is misclassified into the building area in the contrast methods. In contrast, our method produces fewer misclassification. That confirms the fusion features extracted by the CHGFNet have stronger discrimination performance.

2) Classification Results on PoDelta2: In this section, we conduct the experiments on PoDelta2 to observe the performance of the proposed CHGFNet. Two crucial hyperparameters are given: the size of sample patches is 25×25 and the training sample number for each land type is 35 000. The detailed description is shown in Section V-C. Four quantitative metrics of our model and several comparison methods are shown in Table V. Meanwhile, the classification maps are shown in Fig. 16.

Four quantitative metrics of land cover classification are shown in Table V; we discover that the CHGFNet has much better accuracies than the comparison methods. Especially, the CNN-SAR and CNN-Opt. explore single modal for land cover classification, and our model fully captures the complementary information between the optical and SAR features and achieves great performance with 11.95% improvement. Second, in the CNN-MCAM-Pro. and CNN-MCAM-Con., the GHFM is replaced with linear fusion methods (element-wise production and connection) that fuse optical and SAR feature with equal-weight score, so that the performance will decrease. Compared with our model's performance, the PA of CNN-MCAM-Pro. is reduced by 3.8%. In CNN-CBAM-GHFM, the MCAM is replaced with channel attention module of the CBAM [55], which loses the complementary information due to the independent optical channel attention and SAR channel attention. Our proposed model has 1.23% improvement compared with the CNN-CBAM-GHFM. Finally, we compare our method with the state-of-the-art methods, the 2-D-CNN and MRSDC; it is obvious that the CHGFNet achieves a better performance, with 3.95% improvement.

Fig. 16 shows the classification maps of our method and several contrast methods. It is seen that the classification map of our method is smoother and more accurate than other contrast methods. And we enlarge the details with the red and brilliant blue regions to fully compare these land cover maps. In upper right corner area (rod box) of the classification maps of contrast methods, the water area is misclassified into the farmland and tree areas. In contrast, our method produces fewer misclassification. Meanwhile, in lower left corner area

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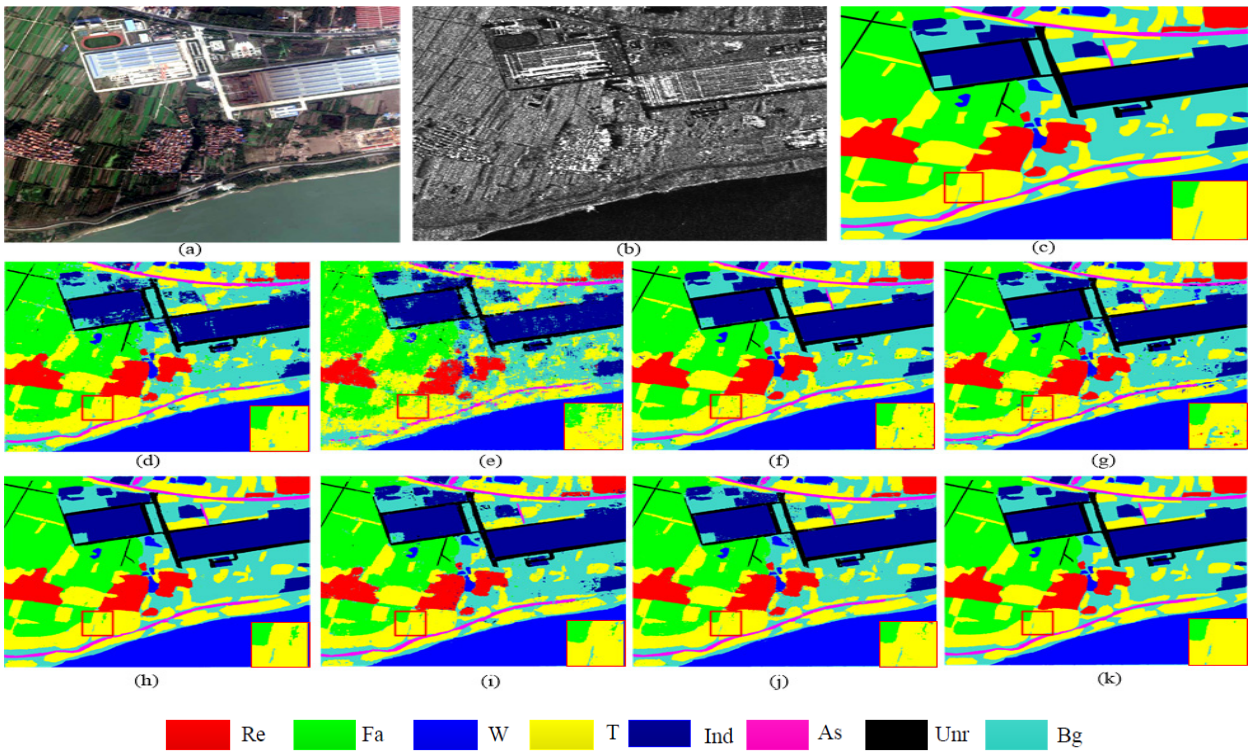


Fig. 17. Classification maps of different methods on WuHan. (a) Original optical image. (b) Original SAR image. (c) Ground truth. (d) CNN-Opt. (e) CNN-SAR. (f) CNN-MCAM-Con. (g) CNN-MCAM-Pro. (h) CNN-CBAM-GHFM. (i) 2-D-CNN. (j) MRSDC. (k) CHGFNet.

(brilliant blue box) of the land cover classification maps of the contrast methods, the unknown class is misclassified into the farmland and tree classes. However, our CHGFNet produces the fewer misclassification. It is seen that our method can efficiently excavate the complementary information between optical and SAR features and obtain the fusion features with the stronger discrimination performance for land cover classification.

3) Classification Results on WuHan: To further verify the abilities of our method, we use a more challenging co-registered optical and SAR data set, called WuHan data set, which includes resident (Re), industrial land (Ind), water (W), tree (T), farmland (Fa), background (Bg), unhardened road (Unr), and asphalt road (As), as shown in Fig. 17. In the experiment of WuHan, two crucial hyperparameters are given: the size of sample patches is 21×21 and the training sample number of each land type is 10 000. The detailed description is shown in Section V-C. To evaluate the performance of our method, Table VI shows four quantitative metrics of our model and several comparison methods, and Fig. 17 presents classification maps of different methods.

Table VI shows the PA, MPA, MIoU, and FWIoU of different methods, and some phenomenon can be observed from four quantitative metrics of land cover classification. First, it is obvious that the CNN-SAR achieves a worst performance compared with the other methods, due to the poor interpretability of SAR images. Second, the multimodal deep learning methods, such as the CNN-MCAM-Pro. and MRSDC, obtain better land cover classification than unimodal deep learning methods, such as the CNN-Opt. and CNN-SAR, which confirm the optical and SAR features have the complementary information to improve the classification accuracies. Third, by comparing our method and variant modals, CNN-MCAM-Con. and CNN-CBAM-GHFM, we can find all the components of our model play an important role for improving performance. The MCAM is explored to capture the semantic interaction between optical and SAR features; GHFM is used to consider the varying contributions of the two modalities for delineating different categories in different scenes. Finally, by comparing the state-of-the-art models, 2-D-CNN and MRSDC, we can know that the CHGFNet achieved a better performance, with 1.48% improvement.

Meanwhile, Fig. 17 shows the classification maps. It is obvious that the classification map of our method is better than other contrast methods. Such as, for asphalt classification maps, the asphalt road derived from our method is more complete and has fewer misclassified. However, the asphalt classification map derived from CNN-SAR is intermittent, and the asphalt classification maps derived from 2-D-CNN MRSDC CNN-MCAM-Con., and CNN-CBAM-GHFM. have some misclassification. Furthermore, we show the details figure in low right corner area (rod box) of Fig. 17, which is used to fully compare these land cover classification maps. We can know that the farmland is misclassified into tree area in contrast methods.

4) Confusion Matrixes: For further analysis, we illustrate the confusion matrix of our method on the PoDelta1, PoDelta2, and WuHan data sets in Fig. 18. From confusion matrixes, the observations can be found: First, all land cover types of PoDelta1, PoDelta2, and WuHan data sets can be well

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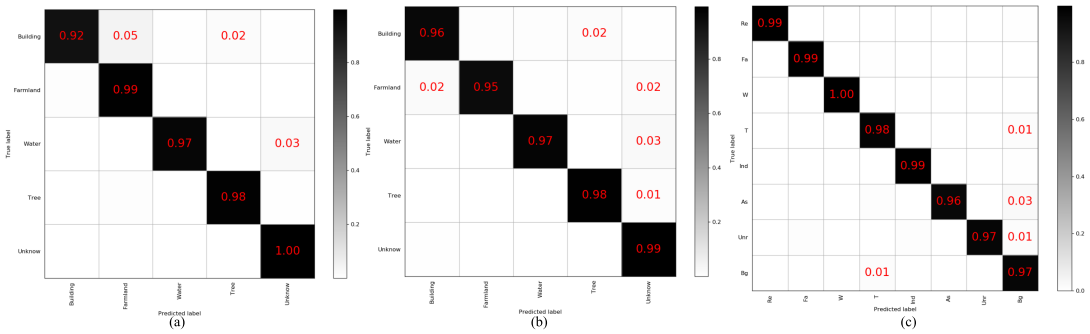


Fig. 18. Confusion matrixes of the proposed CHGFNet. (a) Confusion matrix on the PoDelta1. (b) Confusion matrix on the PoDelta2. (c) Confusion matrix on WuHan.

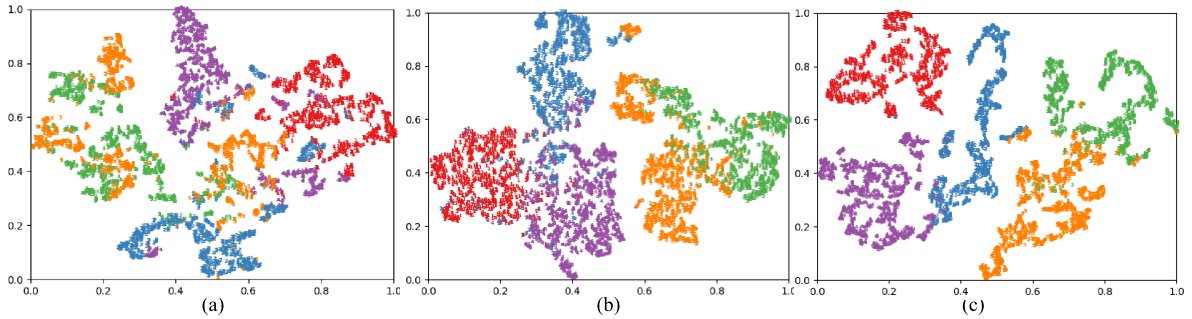


Fig. 19. *t*-SNE visualization on PoDelta1 data set. (a) CNN-SAR. (b) CNN-Opt. (c) CHGFNet. Different colors denote different categories: "red" for building, "blue" for water, "green" for farmland, "purple" for tree, and "yellow" for unknown.

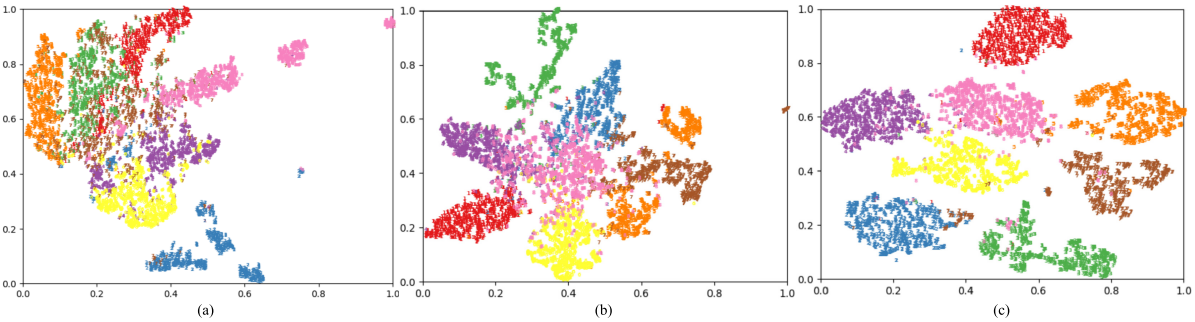


Fig. 20. *t*-SNE visualization on WuHan data set. (a) CNN-SAR. (b) CNN-Opt. (c) CHGFNet. Different colors denote different categories: "red" for Re, "blue" for Fa, "green" for W, "purple" for T, "orange" for Ind, "yellow" for As, "brown" for Unr, and "pink" for Ind.

classified by our method. Meanwhile, for the land cover types with a small number of samples, such as building on PoDelta1, which may contain fewer patterns and structures, building may be misclassified into farmland in Fig. 18(a). In a word, the confusion matrixes prove the superiority of our method.

F. Qualitative Visualization

To verify fusion features extracted by our method have good discrimination, we first accomplish the training of CHGFNet, CNN-Opt., and CNN-SAR (described in Section V-B) based on PoDelta1 and WuHan data sets. Then, we use the *t*-distributed stochastic neighbor embedding (*t*-SNE) [57] to visualize the features from the penultimate fully connected layer of the CHGFNet, CNN-Opt., and CNN-SAR. Figs. 19 and 20 are generated by randomly selecting 1000 matching optical and SAR patch pairs (2000 data points) for each land types, which illustrate the feature distributions of different land types on the PoDelta1 and WuHan data sets using different colors in the 2-D plane, respectively. From Fig. 19(a), we can see that the features distributions of different land types are significantly overlapped in CNN-SAR, which means that different land types are hard to distinguish. Fig. 19(b) shows the features distributions of different land types in CNN-Opt., we can see that the interclass distance in different land types is small, and the features distributions of different land types have a degree of overlap. Furthermore, the boundaries of different land types for the CHGFNet in Fig. 19(c) are clearer than those in Fig. 19(a) and (b). In addition, there is a large interclass distance in each land types. Similarly, from Fig. 20, we can find the interclass distances in Fig. 20(c) are more than Fig. 20(a) and (b).

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To summarize, these results indicate that the fusion features extracted by the CHGFNet show strong discriminative ability for all land types.

VI. Conclusion

In this article, we show the distribution histograms of deep semantic features extracted from optical and SAR modalities within land categories, which intuitively demonstrates there is the large complementary potential between the optical and SAR features extracted by CNNs. Thus, we propose a novel optical and SAR feature fusion model, called the CHGFNet, which hierarchically fuses both optical and SAR features to focus on their complementary for land cover classification. More specifically, CHGFNet consists of three main components: a two-stream feature extractor, the MCAM, and the GHFM. Given the optical and SAR patch pairs, a two-stream feature extractor introduces a multistage feature learning methodology to acquire the optical and SAR features with the discriminative capabilities, respectively. Then, the MCAM is embedded into the CHGFNet, which provides an efficient stage to capture the correlation between the optical and SAR features by jointly computing collaborative attention response in joint feature space, which is used to explore the complementarity between optical and SAR features. Finally, the GHFM automatically achieves the weighted fusion of both optical and SAR features for classifying different land cover categories. Extensive comparative evaluations demonstrate the advantages of the CHGFNet on three broad co-registered optical and SAR data sets. Moreover, how to realize the automatic setting of parameters (e.g., the initial learning rate, batch size, and epoch) is an interesting research point in future, which makes this work can be used or developed by other researchers.

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Xiao Li received the M.S. degree in control science and engineering from Xiangtan University, Xiangtan, China, in 2018. He is pursuing the Ph.D. degree in information and communication engineering with the National University of Defense Technology, Changsha, China.

His research interests include image processing and pattern recognition, sparse representation, deep learning, and remote sensing image applications.



Lin Lei was born in New York, NY, USA, in 1977. She received the B.S. and M.S. degrees and the Ph.D. degree in information and communication engineering from the National University of Defense Technology, Changsha, China, in 2008.

She is an Associate Professor with the School of Electronic Science, National University of Defense Technology. Her research interests include computer vision, remote sensing image interpretation, and data fusion.



Yuli Sun received the M.S. degree from the University of Science and Technology of China, Hefei, China, in 2014. He is pursuing the Ph.D. degree with the College of Electronic Science, National University of Defense Technology, Changsha, China.

His research interests cover machine learning and remote sensing image processing.



Ming Li received the bachelor's and M.S. degrees from Central South University, Changsha, China, in 2013 and 2017, respectively. He is pursuing the Ph.D. degree with the College of Electronic Science, National University of Defense Technology, Changsha.

His research interests cover remote sensing object detection, image retrieval, and change detection.



Guangyao Kuang (Senior Member, IEEE) received the B.S. and M.S. degrees in geophysics from the Central South University of Technology, Changsha, China, in 1988 and 1991, respectively, and the Ph.D. degree in communication and information from the National University of Defense Technology, Changsha, in 1995.

He is a Professor with the School of Electronic Science, National University of Defense Technology. His research interests include remote sensing, synthetic aperture radar (SAR) image processing, change detection, SAR ground moving target indication, and classification with polarimetric SAR images.

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